

Advanced Excel Formula Practice Set

Table 1: Employee Performance & Incentive Report

Emp ID	Employee Name	Department	Salary	Monthly Sales	Attendance %	Performance Rating
E101	Amit	Sales	55000	85000	92	Excellent
E102	Neha	HR	42000	30000	88	Good
E103	Ravi	IT	65000	95000	96	Excellent
E104	Simran	Sales	48000	72000	85	Average
E105	Karan	IT	70000	105000	98	Excellent
E106	Priya	HR	39000	25000	91	Good
E107	Mohit	Sales	52000	68000	89	Good
E108	Pooja	IT	61000	88000	94	Excellent

Questions

1.

The management wants to identify employees eligible for a quarterly performance bonus. An employee will be considered eligible only if the employee has achieved sales greater than ₹80,000 and has attendance above 90%. However, employees with an “Excellent” performance rating should also be considered eligible even if one of the above conditions is not satisfied. Write a formula using **IF**, **AND**, and **OR** to display either “Bonus Eligible” or “Not Eligible”.

2.

The HR department wants to classify employees based on their salary structure. Create a formula that displays:

- “High Income Group” if salary is greater than ₹65,000

- “Medium Income Group” if salary is between ₹50,000 and ₹65,000
- “Basic Income Group” for all remaining employees.

Use nested **IF** statements.

3.

The finance department wants to calculate the total salary expense for employees working in the IT department only. Use an appropriate **SUMIF** formula to calculate the total amount paid to the IT department employees.

4.

The company wants to know how many employees have attendance greater than 90% because those employees are eligible for an attendance reward program. Use **COUNTIF** formula to find the total number of such employees.

5.

The sales manager wants to calculate the average monthly sales of employees who achieved sales above ₹75,000. Use **AVERAGEIF** formula to determine the required average.

6.

Create a formula to identify “Top Performers” in the organization. An employee should be marked as a “Top Performer” if:

- Salary is greater than ₹60,000
- AND sales are above ₹90,000
- OR performance rating is “Excellent” with attendance above 95%.

Use a combination of **IF**, **AND**, and **OR**.

7.

The company wants to categorize employee performance according to monthly sales:

- Sales above ₹1,00,000 → “Outstanding”
- Sales between ₹75,000 and ₹1,00,000 → “Very Good”
- Sales between ₹50,000 and ₹75,000 → “Good”
- Below ₹50,000 → “Needs Improvement”

Use nested IF statements.

8.

The HR manager wants to calculate the total number of employees whose performance rating is “Excellent”. Use COUNTIF formula for the calculation and display the result in a separate summary section.

Table 2: Student Academic Performance Report

Roll No	Student Name	Classes	English	Maths	Science	Computer	Attendance %
101	Rahul	12A	85	92	88	91	95
102	Priya	12A	72	65	70	68	89
103	Aman	12B	90	96	94	98	98
104	Neha	12B	60	58	62	55	80
105	Simran	12A	78	81	75	84	91
106	Mohit	12B	88	73	85	79	93
107	Karan	12A	55	49	60	52	76
108	Pooja	12B	92	95	97	99	99

Questions

1.

Calculate the total marks obtained by each student in all four subjects. Then calculate the percentage assuming the maximum marks are 400. Use suitable Excel formulas.

2.

The school wants to award scholarships to deserving students. A student will receive a scholarship if:

- Percentage is greater than 85%
- AND attendance is above 90%
- OR Computer marks are above 95.

Use **IF**, **AND**, and **OR** formulas to display “Scholarship Approved” or “Not Approved”.

3.

Create a grading system using nested **IF** statements:

- Percentage ≥ 90 → Grade A+
 - Percentage ≥ 75 → Grade A
 - Percentage ≥ 60 → Grade B
 - Percentage ≥ 40 → Grade C
 - Below 40 → Fail
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4.

The principal wants to know the average marks in Maths for students who scored more than 70 marks in Maths. Use **AVERAGEIF** formula.

5.

The administration department wants to calculate the total English marks of students belonging to class 12A only. Use **SUMIF** formula.

6.

The school wants to identify weak students. If a student has scored below 50 in any one subject, the result should display “Needs Improvement”; otherwise display “Satisfactory”. Use **OR** function with **IF**.

7.

Count the number of students whose attendance percentage is greater than 90 because they are eligible for the “Best Attendance Award”. Use **COUNTIF**.

8.

Create a formula to identify “Academic Excellence” students. A student should receive this title if:

- English, Maths, Science, and Computer marks are all above 85
- AND attendance is above 90%.

Use **AND** with **IF**.

Table 3: Product Inventory & Sales Database

Product ID	Product Name	Category	Price	Stock Available
P101	Keyboard	Electronics	800	50
P102	Mouse	Electronics	500	120

P103	Printer	Electronics	9500	15
P104	Chair	Furniture	3500	25
P105	Desk	Furniture	7000	10
P106	Pen Drive	Accessories	650	80
P107	Router	Electronics	2500	18
P108	Notebook	Stationery	120	200

Questions

1.

The sales department wants to automatically retrieve the product name and product price whenever a Product ID is entered into a separate cell. Use **VLOOKUP** formula to fetch both details from the table.

2.

The inventory manager wants to identify products that require urgent restocking. If stock available is less than 20, the result should display "Restock Immediately"; otherwise display "Stock Sufficient". Use **IF** formula.

3.

Calculate the total stock available for all products belonging to the Electronics category using **SUMIF**.

4.

The company wants to know how many products currently have stock greater than 50 units. Use **COUNTIF** formula.

5.

Calculate the average price of products whose price is greater than ₹1,000 using **AVERAGEIF**.

6.

The management wants to identify fast-selling products. A product should be marked as “Fast Moving” if:

- Stock is less than 30
- AND price is greater than ₹2,000
- OR category is Electronics.

Use **IF**, **AND**, and **OR** formulas.

7.

Create a horizontal sales table and use **HLOOKUP** formula to retrieve monthly sales figures for a selected category.

Category	Jan	Feb	Mar	Apr
Electronics	45000	52000	61000	58000

Retrieve March sales using **HLOOKUP**.

8.

The store manager wants to classify products according to pricing:

- Above ₹7,000 → “Premium Product”
- ₹2,000 to ₹7,000 → “Standard Product”
- Below ₹2,000 → “Budget Product”

Use nested **IF** statements to generate the classification automatically.

Table 4:

Table: Employee Promotion & Incentive Report

Emp ID	Employee Name	Department	Salary	Sales	Attendance %	Rating	Experience
E301	Amit	Sales	65000	120000	96	Excellent	6
E302	Priya	HR	48000	45000	89	Good	3
E303	Rohit	IT	72000	98000	94	Excellent	7
E304	Simran	Sales	55000	76000	91	Average	4
E305	Karan	IT	81000	135000	98	Excellent	9
E306	Neha	HR	43000	39000	93	Good	5
E307	Mohit	Sales	59000	68000	87	Good	2
E308	Pooja	IT	70000	112000	97	Excellent	8

Hard Level Tasks

1.

The company wants to identify employees eligible for an international business trip. An employee will qualify only if:

- Sales are greater than ₹1,00,000 AND attendance is greater than 95%
- OR the employee belongs to the IT department AND has more than 7 years of experience.

Create a formula using **IF**, **AND**, and **OR** to display “Eligible for Trip” or “Not Eligible”.

2.

The HR department wants to approve leadership promotions only for employees who satisfy the following conditions:

- Salary greater than ₹70,000 AND performance rating is “Excellent” AND attendance above 90%
- OR sales are greater than ₹1,30,000 regardless of department.

Use a combination of **IF**, **AND**, and **OR** formulas to display “Promotion Approved” or “Promotion Rejected”.

3.

The company wants to identify employees who should receive a “Critical Employee Retention Bonus”. The conditions are:

- Experience greater than 5 years AND salary below ₹75,000
- OR sales above ₹1,10,000 AND rating is “Excellent”.

Write a formula using **IF**, **AND**, and **OR**.

4.

The finance department wants to mark employees as “High Risk Employees” if:

- Attendance is below 90% AND sales are below ₹70,000
- OR rating is “Average”
- OR experience is less than 3 years.

Use advanced logical formulas to generate the required result.

5.

The management wants to shortlist employees for executive training. An employee should be selected only if:

- Department is IT AND attendance above 95% AND sales above ₹90,000
- OR department is Sales AND sales above ₹1,20,000 AND rating is “Excellent”.

Use nested **AND** and **OR** formulas with **IF**.

6.

The company wants to provide a special “Loyalty Appreciation Award” to employees who satisfy any one of the following conditions:

- Experience greater than 8 years AND attendance greater than 95%
- OR sales above ₹1,00,000 AND salary less than ₹80,000
- OR rating is “Excellent” AND department is IT.

Create a hard-level logical formula using multiple **AND** and **OR** conditions.

7.

The HR manager wants to identify employees who require performance counseling. An employee should be marked for counseling if:

- Attendance below 90% AND sales below ₹80,000
- OR rating is “Average”
- OR salary below ₹50,000 AND experience below 4 years.

Use **IF**, **AND**, and **OR** formulas.

8.

The company wants to identify employees who qualify for the “Elite Performer Club”. The eligibility criteria are:

- Sales greater than ₹1,00,000 AND attendance above 95% AND rating is “Excellent”
- OR experience greater than 8 years AND salary greater than ₹80,000.